

University Preparatory Program (Mathematics) Infinite Series



Engineering Mathematics Team

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3. Infinite Series

Learning Objectives

Upon completion of this lesson, you should be able to:

1. Test the convergence of an infinite series using Ratio Test or Comparison Test
2. Define power series and its interval of convergence
3. Compute the interval of convergence of a power series
4. Construct Taylor/Maclaurin series of a function

3.1 Series and Convergence

To understand the concept of a series and its convergence, we begin with the discussion of sequences.

Definition 3.1.1 (Sequence and Convergence)

A **sequence** f is a function with domain $\mathbb{N}$, the set of all positive integers $\{1, 2, 3, \dots\}$. The notation f_n is often used to denote $f(n)$. A sequence f is **convergent** if its limit exists, i.e.

$$\lim_{n \rightarrow \infty} f_n = L < \infty$$

We often denote a sequence f by $(f_n)_{n=1}^{\infty}$. To find the limit of a sequence, we may apply the rules and theorems for limits of a function.

A sequence f is **bounded** if there is a positive number A such that $|f_n| \leq A$ for all $n \in \mathbb{N}$; it is **monotonic** if it is either:

- strictly increasing: $f_n < f_{n+1}$ for all $n \in \mathbb{N}$, or
- increasing: $f_n \leq f_{n+1}$ for all $n \in \mathbb{N}$, or
- strictly decreasing: $f_n > f_{n+1}$ for all $n \in \mathbb{N}$, or
- decreasing: $f_n \geq f_{n+1}$ for all $n \in \mathbb{N}$.

Theorem 3.1.2 (Bounded Monotonic Convergence Theorem, BMCT)

A bounded and monotonic sequence is convergent.

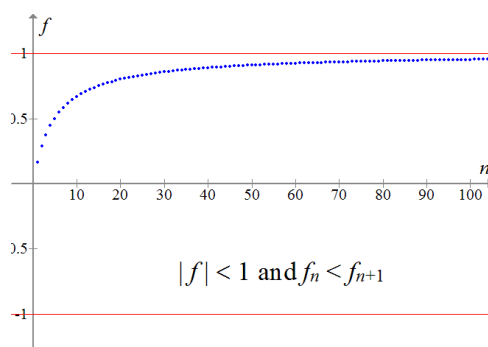


Figure 3.1.1. A bounded and monotonic sequence

Example 3.1.3

- (a) For $0 < r < 1$, the power sequence $(r^n)_{n=1}^{\infty}$ is convergent; it is bounded because $|r^n| < 1$, and monotonic because $r^{n+1} < r^n$ for all $n \in \mathbb{N}$.
- (b) The alternating sequence $((-1)^n)_{n=1}^{\infty}$ is divergent; it is bounded but not monotonic.
- (c) The sequence $(2^n)_{n=1}^{\infty}$ is divergent; it is monotonic but not bounded.

Note: there are convergent sequences which are bounded but not monotonic, e.g. $(r^n)_{n=1}^{\infty}$ for $-1 < r < 0$.

Definition 3.1.4 (Series and Convergence)

An infinite **series** is a summation

$$\sum_{k=1}^{\infty} a_k$$

A series is **convergent** if the sequence of its partial sums $S_n = \sum_{k=1}^n a_k$ is convergent, that is, $\lim_{n \rightarrow \infty} S_n$ exists; in this case, this limit is the value of the infinite sum.

Here are examples of some common series and their convergence.

Example 3.1.5

- (a) The **geometric series** $\sum_{k=0}^{\infty} r^k$ is convergent for $|r| < 1$ and divergent for $|r| \geq 1$.
- (b) The **harmonic series** $\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent.
- (c) The **p-series** $\sum_{k=1}^{\infty} \frac{1}{k^p}$ is convergent for $p > 1$ and divergent for $0 < p \leq 1$.

3.2 Testing the Convergence of a Series

In this section, we will discuss some techniques to determine the convergence of a given series:

1. Divergence Test
2. Ratio Test
3. Comparison Test

The Divergent Test is a simple test to conclude that a given series is NOT convergent, i.e. it is divergent. It is based on the following theorem.

Theorem 3.2.1

If the series $\sum_{k=1}^{\infty} a_k$ is convergent, then $\lim_{k \rightarrow \infty} a_k = 0$.

Proof.

Let $S = \sum_{k=1}^{\infty} a_k$ and $S_n = \sum_{k=1}^n a_k$. Then $\lim_{n \rightarrow \infty} S_n = S$ and $\lim_{n \rightarrow \infty} S_{n-1} = S$. Since $a_n = S_n - S_{n-1}$, we have $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (S_n - S_{n-1}) = S - S = 0$. ■

An immediate consequence of Theorem 2.2.1 is the Divergence Test.

Theorem 3.2.2 (Divergence Test)

If $\lim_{k \rightarrow \infty} a_k \neq 0$, then the series $\sum_{k=1}^{\infty} a_k$ is divergent.

Note that the Divergence Test is inconclusive if $\lim_{k \rightarrow \infty} a_k = 0$. In this case, we need to use a different test.

Example 3.2.3

Determine the convergence of each of the following series.

(a) $\sum_{k=1}^{\infty} \frac{k}{k+1}$

(b) $\sum_{k=1}^{\infty} \frac{k}{e^k}$

(a) $\lim_{k \rightarrow \infty} \frac{k}{k+1} = \lim_{k \rightarrow \infty} \left(1 - \frac{1}{k+1}\right) = 1 \Rightarrow \sum_{k=1}^{\infty} \frac{k}{k+1}$ is divergent.

(b) $\lim_{k \rightarrow \infty} \frac{k}{e^k} = \lim_{k \rightarrow \infty} \frac{1}{e^k} = 0 \Rightarrow$ the divergent test is inconclusive.

The Divergence Test fails to conclude whether the series in Example 2.2.3(b) is divergent. To investigate the convergence of such series, we need to use a different test, e.g. Ratio Test.

Theorem 3.2.4 (Ratio Test)

Given a series $\sum_{k=1}^{\infty} a_k$, let $L = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right|$. The series is convergent if $L < 1$, and divergent if $L > 1$ or $L = \infty$.

Note that the ratio test is inconclusive if $L = 1$. In this case, we need to use a different test.

Example 3.2.5

Determine the convergence of each of the following series.

(a) $\sum_{k=1}^{\infty} \frac{k}{e^k}$

(b) $\sum_{k=0}^{\infty} \frac{k!}{e^k}$

(c) $\sum_{k=1}^{\infty} \frac{k}{k^2 + 1}$

$$\begin{aligned} \text{(a)} \quad \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| &= \lim_{k \rightarrow \infty} \left| \frac{k+1}{e^{k+1}} \times \frac{e^k}{k} \right| = \lim_{k \rightarrow \infty} \frac{k+1}{ek} = \frac{1}{e} \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k} \right) = \frac{1}{e} < 1 \\ &\Rightarrow \sum_{k=1}^{\infty} \frac{k}{e^k} \text{ is convergent.} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| &= \lim_{k \rightarrow \infty} \left| \frac{(k+1)!}{e^{k+1}} \times \frac{e^k}{k!} \right| = \lim_{k \rightarrow \infty} \frac{(k+1) \cdot k!}{e \cdot k!} = \frac{1}{e} \lim_{k \rightarrow \infty} (k+1) = \infty \\ &\Rightarrow \sum_{k=0}^{\infty} \frac{k!}{e^k} \text{ is divergent.} \end{aligned}$$

$$\text{(c)} \quad \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{k+1}{(k+1)^2 + 1} \times \frac{k^2 + 1}{k} \right| = \lim_{k \rightarrow \infty} \frac{k^3 + k^2 + k + 1}{k^3 + 2k^2 + 2k} = 1$$

In this case, the Ratio Test is inconclusive.

Both The Divergence and Ratio Tests fail to determine the convergence of the series in Example 3.2.5(c). To investigate the convergence of such series, we need to use a different test, e.g. Comparison Test.

Theorem 3.2.6 (Comparison Test)

Given a convergent series $\sum_{k=1}^{\infty} c_k$ and a divergent series $\sum_{k=1}^{\infty} d_k$. The series $\sum_{k=1}^{\infty} a_k$ (where $a_k \geq 0$) is

1. convergent if $a_k \leq c_k$ for all $k \geq m$ (where $m \in \mathbb{N}$)
2. divergent if $0 \leq d_k \leq a_k$ for all $k \geq m$ (where $m \in \mathbb{N}$)

To use the comparison test, we usually compare the given series with another series (e.g. geometric series or p-series) for which the convergence is known.

Example 3.2.7

Determine the convergence of each of the following series.

(a) $\sum_{k=1}^{\infty} \frac{k}{k^2 + 1}$

(b) $\sum_{k=0}^{\infty} \frac{1}{\sqrt{k+2}}$

(c) $\sum_{k=1}^{\infty} \frac{1}{k^k}$

(a) Comparing $\frac{k}{k^2 + 1}$ with $\frac{1}{2k}$, we observe that $\frac{1}{2k} \leq \frac{k}{k^2 + 1}$ for all $k \in \mathbb{N}$.

Since $\sum_{k=1}^{\infty} \frac{1}{2k}$ (harmonic series) is divergent, the series $\sum_{k=1}^{\infty} \frac{k}{k^2 + 1}$ is divergent.

(b) Comparing $\frac{1}{\sqrt{k+2}}$ with $\frac{1}{\sqrt{2k}}$, we observe that $\frac{1}{\sqrt{2k}} \leq \frac{1}{\sqrt{k+2}}$ for all $k \geq 2$.

Since $\sum_{k=1}^{\infty} \frac{1}{\sqrt{2k}}$ (p-series with $p = \frac{1}{2}$) is divergent, the series $\sum_{k=0}^{\infty} \frac{1}{\sqrt{k+2}}$ is divergent.

(c) Comparing $\frac{1}{k^k}$ with $\frac{1}{2^k}$, we observe that $\frac{1}{k^k} \leq \frac{1}{2^k}$ for all $k \geq 2$.

Since $\sum_{k=1}^{\infty} \frac{1}{2^k}$ (geometric series with $r = \frac{1}{2}$) is convergent, the series $\sum_{k=1}^{\infty} \frac{1}{k^k}$ is convergent.

A series with alternate signs on its terms is called an **alternating series**. The alternate signs are denoted by $(-1)^k$ or $(-1)^{k+1}$ or $(-1)^{k-1}$. The following are examples of alternating series. The convergence test for alternating series is omitted in our discussion.

Example 3.2.8

(a) The **alternating p-series** $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^p}$ is convergent for $p > 0$.

(b) The series $\sum_{k=1}^{\infty} (-1)^k k$ is divergent.

The convergent of alternating harmonic series will be very useful in finding the interval of convergence of a power series, which is discussed in the next section.

3.3 Power Series

Definition 3.3.1 (Power Series)

A **power series** in $(x - c)$ is an infinite series of the form $\sum_{k=0}^{\infty} a_k (x - c)^k$.

A power series is a function in x ; its domain consists of the values for which the series converges. The following theorem states three cases for the convergence of a power series.

Theorem 3.3.2 (Convergence of a Power Series)

Given a power series $\sum_{k=0}^{\infty} a_k x^k$.

Then exactly one of the following is true:

1. The series converges only for $x = 0$
2. The series converges for all x
3. The series converges *absolutely* for all $|x| < R$ and diverges for $|x| > R$, where $R > 0$ is the **radius of convergence** of the series; it may either converge or diverge at the endpoints $x = -R$ and $x = R$

In the trivial case, i.e. where the power series converges only for $x = 0$, its radius of convergence is zero. And in the case where the power series converges for all x , its radius of convergence is infinite.

The **interval of convergence** of a power series is an interval on which the power series is convergent; it can be found by using ratio test. The following example demonstrates this procedure.

Example 3.3.3

Determine the interval of convergence of the power series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (x-1)^k$.

By the Ratio Test, the series is convergent if $\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1$.

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(-1)^{k+2} (x-1)^{k+1}}{k+1} \times \frac{k}{(-1)^{k+1} (x-1)^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{k}{k+1} (x-1) \right| = |x-1|.$$

So,

$$\begin{aligned} \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 &\Rightarrow |x-1| < 1 \\ &\Rightarrow -1 < x-1 < 1 \\ &\Rightarrow 0 < x < 2. \end{aligned}$$

Now we observe the convergence at the endpoints: $x = 0$ and $x = 2$.

$$x = 0 \Rightarrow \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (-1)^k = \sum_{k=1}^{\infty} \frac{(-1)^{2k+1}}{k} = - \sum_{k=1}^{\infty} \frac{1}{k} \text{ is a divergent series.}$$

$$x = 2 \Rightarrow \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (1)^k = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \text{ is a convergent series, cf. Example 3.2.8(a).}$$

Thus, the interval of convergence for the power series $\sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{k} (x-1)^k$ is $0 < x \leq 2$.

As a function, power series provides a very simple calculation to find its derivative or integral.

Theorem 3.3.4 (Derivative and Integral of a Power Series)

Given a power series $f(x) = \sum_{k=0}^{\infty} a_k x^k$ with radius of convergence $R > 0$. Then for $|x| < R$,

$$f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$$

and

$$\int f(x) dx = C + \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}$$

Note to Theorem 3.3.4: if a particular antiderivative $F(x)$ of $f(x)$ is determined, then $C = F(0)$.

Example 3.3.5

Given that $\frac{1}{1+x} = 1 - x + x^2 - x^3 \dots = \sum_{k=0}^{\infty} (-1)^k x^k$ for $|x| < 1$, find the power series

representation of each of the following functions for $|x| < 1$.

(a) $f(x) = \frac{1}{(1+x)^2}$

(b) $g(x) = \ln(1+x)$

(a) Since $\frac{1}{(1+x)^2} = -\frac{d}{dx} \left(\frac{1}{1+x} \right)$,

$$\frac{1}{(1+x)^2} = -\sum_{k=0}^{\infty} (-1)^k \frac{d}{dx} (x^k) = \sum_{k=1}^{\infty} (-1)^{k+1} k x^{k-1} = 1 - 2x + 3x^2 - 4x^3 + \dots$$

(b) Since $\ln(1+x) = \int \frac{1}{1+x} dx$,

$$\ln(1+x) = \sum_{k=0}^{\infty} (-1)^k \left(\int x^k dx \right) = C + \sum_{k=0}^{\infty} \frac{(-1)^k}{k+1} x^{k+1} = C + x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

where $C = g(0) = \ln 1 = 0$. Thus, $\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$

3.4 Taylor and Maclaurin Series

A function f is an **infinitely differentiable** function on an interval I if all its derivatives exist on I . Many of elementary and trigonometric functions are infinitely differentiable functions. Power series can be used to represent such functions on a certain interval.

Theorem 3.4.1 (Uniqueness for Power Series Representation)

Let c be a fixed number and f be an infinitely differentiable function on the interval $(c-R, c+R)$ for some $R > 0$. If f has a power series representation

$$f(x) = \sum_{k=0}^{\infty} a_k (x-c)^k \text{ for all } x \in (c-R, c+R),$$

then the representation is unique and the coefficients a_k is given by

$$a_k = \frac{f^{(k)}(c)}{k!} \text{ for } k = 0, 1, 2, \dots$$

The power series representation above is called the **Taylor series** of f at $x = c$. When $c = 0$, the power series is called the **Maclaurin series** of f . The following are Taylor/Maclaurin series for elementary and trigonometric functions.

Example 3.4.2

$$(a) \quad \frac{1}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^k \quad \text{for } -1 < x < 1$$

$$(b) \quad e^x = \sum_{k=0}^{\infty} \frac{1}{k!} x^k \quad \text{for all } x$$

$$(c) \quad \ln x = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (x-1)^k \quad \text{for } 0 < x \leq 2$$

$$(d) \quad \sin x = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{(2k-1)!} x^{2k-1} \quad \text{for all } x$$

$$(e) \quad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \quad \text{for all } x$$

The Taylor series of a function f at $x = c$ may not represent the function on its domain. For instance, the Taylor series of $\ln x$ at $x = 1$ converges only on the interval $0 < x \leq 2$, even though $\ln x$ is defined for $x > 0$, as shown in Figure 3.4.1.

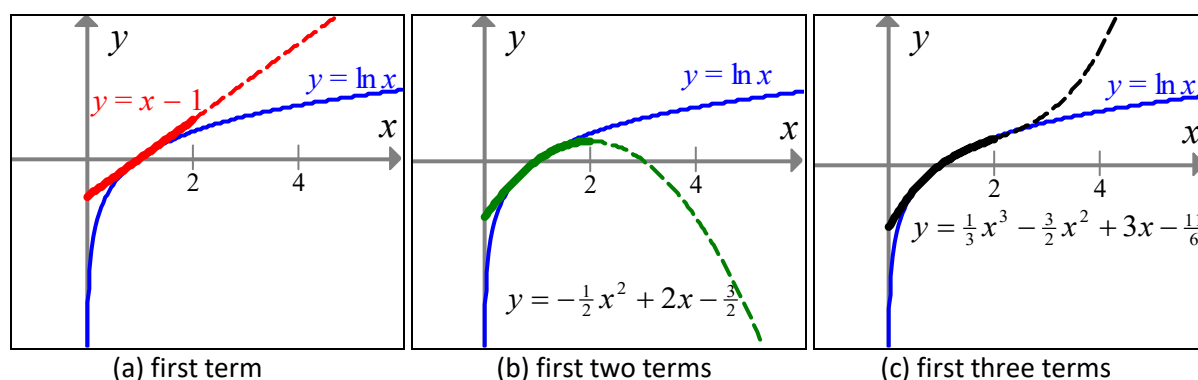


Figure 3.4.1. Various partial sums of the Taylor series of $\ln x$ at $x = 1$

3.5 Exercises

1. Determine the convergence of each of the following series.

(a) $\sum_{k=0}^{\infty} \left(\frac{e}{\pi}\right)^k$

(d) $\sum_{k=2}^{\infty} \frac{1}{\sqrt{k}-1}$

(b) $\sum_{k=1}^{\infty} \frac{2}{k\sqrt{k}}$

(e) $\sum_{k=2}^{\infty} \frac{(k-1)!}{(k+1)^2}$

(c) $\sum_{k=1}^{\infty} \frac{1}{k2^k}$

(f) $\sum_{k=2}^{\infty} \frac{2^{k+2}}{\ln k}$

2. Let $|r| < 1$. By considering the partial sums S_n and S_{n-1} , show that $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$.

3. Show that $\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent by comparing it with the improper integral $\int_1^{\infty} \frac{1}{x} dx$.

4. Find the interval of convergence of each of the following power series.

(a) $\sum_{k=1}^{\infty} \frac{(x-1)^k}{\sqrt{k}}$

(e) $\sum_{k=0}^{\infty} \frac{(x-1)^k}{5^k}$

(b) $\sum_{k=1}^{\infty} \frac{2^k x^{2k}}{k}$

(f) $\sum_{k=0}^{\infty} \frac{(x+1)^{2k}}{4^k}$

(c) $\sum_{k=1}^{\infty} \frac{(2x-1)^{2k+1}}{k^{3/2}}$

(g) $\sum_{k=2}^{\infty} \frac{(-2)^k x^k}{\sqrt{k}-1}$

(d) $\sum_{k=1}^{\infty} \frac{x^k}{5^k \sqrt{k}}$

(h) $\sum_{k=1}^{\infty} \frac{(-1)^k (x-1)^k}{k^2}$

5. Referring to the functions given in Example 3.4.2,
 (a) Use Theorem 3.4.1 to derive the Taylor/Maclaurin series formula for each function.
 (b) Determine the interval of convergence for each power series.
6. Use term-by-term differentiation/integration and appropriate Taylor/Maclaurin series to find the power series representation for $f(x) = \tan^{-1} x$.

